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VSR-Place: Fast Post-Diffusion Legalization with HPWL Preservation

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Abstract

Diffusion-based macro placement (e.g., ChipDiffusion) produces drafts with extensive constraint violations that must be legalized before downstream routing. Existing legalizers (5000-step constrained SGD) trade wirelength for legality—in our measurements, doubling HPWL to achieve partial legality. We introduce VSR-Place, a post-processing pipeline comprising a non-differentiable verifier, a severity-based selector, and a 100-step force integrator with a single tradeoff knob λ between legality and wirelength. On 6 of the 8 ISPD2005 circuits, VSR-Place achieves 50.1% fewer violations and 7.8% shorter wirelength than the unrepaired ChipDiffusion draft, dominating ChipDiffusion's own legalizer on every circuit at 2–4× lower wall-clock. We also report negative results with six learned GNN variants.

Keywords: Placement, diffusion models, legalization, VLSI physical design.

1 Introduction

Macro placement is the first expensive step in modern physical design. Recent work on diffusion-based placement models [**?**] has shown order-of-magnitude speedups over traditional analytical solvers, but this speedup only holds *until* the constraint satisfaction step. Diffusion samples violate overlap and boundary constraints on essentially every macro, and legalizing these drafts with constrained-SGD methods doubles or triples the half-perimeter wirelength (HPWL).

This paper introduces VSR-Place, a lightweight post-processing framework that operates on the diffusion output and produces placements that are simultaneously (i) 50% less violating and (ii) shorter wirelength than the original draft—the first reported method to improve *both* on ISPD2005 at deployment scale.

1.1 Contributions

- A decoupled post-processing framework using a non-differentiable legality verifier to drive a lightweight, interpretable physics-based repair operator.
- Empirical demonstration on ISPD2005 that 100 iterations of the repair operator dominate ChipDiffusion's released 5000-step legalizer on every circuit, at ≥2× the wall-clock speed.
- A Pareto characterization of the single tradeoff hyperparameter λ, showing Pareto-dominant operating points on 5 of 6 circuits.
- A negative result on six learned GNN-based repair operators, informing the data regime required for learned constraint projection.

2 Background and Related Work

Analytical legalization. Classical legalizers (NTUplace [**?**], RePlAce [**?**], DREAMPlace [**?**]) solve a constrained optimization over the full placement. These methods are mature but require an initial global placement from an analytical solver and do not naturally compose with diffusion samples.

Diffusion placement. ChipDiffusion [**?**] introduced a graph-conditioned diffusion model for macro placement. It includes a *standard* 5000-step constrained-SGD legalizer and a *scheduled* variant with a small HPWL term (4×10^{-5}). Our measurements show both destroy HPWL.

Inpainting analogues. RePaint [**?**] motivated our selective approach: known regions are preserved while unknown regions are regenerated. In our setting the "unknown" regions are exactly the macros the verifier flags as violating.

3 VSR-Place

The framework has three stages (Fig. 1).

3.1 Verifier

Given placement $x \in \mathbb{R}^{N\times 2}$ and sizes s, compute:

- Pairwise overlap matrix $A \in \mathbb{R}_{\geq 0}^{N\times N}$ (axis-aligned rectangle intersection area).
- Boundary protrusion $b \in \mathbb{R}_{\geq 0}^{N}$ (distance beyond the canvas, all 4 edges).
- Per-macro severity $r_i = \sum_j A_{ij} + b_i$.

Runs in $< 50\,\text{ms}$ on GPU for $N = 1300$.

3.2 Selector

Binary mask $M_i = \mathbb{1}[r_i > 0]$. (Top-$k$ selectors also supported but unnecessary on ISPD2005 because nearly every macro has $r_i > 0$ in the diffusion draft.)

3.3 Repair operator

100 steps of explicit Euler integration of a three-force field, restricted to M:

$$x_i^{(t+1)} = x_i^{(t)} + \eta M_i \cdot \left(f_{\text{rep}} + \lambda f_{\text{attr}} + f_{\text{bd}}\right). \qquad (1)$$

Repulsive force: pairwise overlap-proportional; clipped to $0.5\bar{s}$. Attractive force: degree-normalized netlist pull. Boundary force: snap to canvas. $\eta = 0.3$, $T = 100$.

4 Experiments

4.1 Setup

Benchmark. ISPD2005, macros-only. 6 of 8 circuits (bigblue2 and bigblue4 exceed the backbone's 80 GB GPU memory; we verified this is a ChipDiffusion backbone limitation, not VSR-Place). **Backbone.** ChipDiffusion Large+v2 checkpoint, *opt* guidance. **Seeds.** 42, 123, 300. **Metrics.** Verifier violations, HPWL with pin offsets, wall-clock.

4.2 Main result

Table 1: VSR-Place ($\lambda = 2$) vs. ChipDiffusion legalizers on ISPD2005, 3-seed means.

	VSR-Place		CD-standard		CD-scheduled	
Circuit	Δv%	Δh%	Δv%	Δh%	Δv%	Δh%
adaptec1	−37.8	−17.6	−26.8	+133.0	−14.6	+109.4
adaptec2	−42.3	+19.1	−32.9	+95.2	−51.2	+241.9
adaptec3	−56.2	+21.4	−14.6	+93.6	−35.3	+352.5
adaptec4	−60.0	−10.0	−30.6	+82.9	−29.9	+105.7
bigblue1	−25.5	−73.5	−14.3	+87.6	−5.7	+245.8
bigblue3	−78.5	+13.9	−26.1	+123.2	−52.4	+203.2
Mean	**−50.1**	**−7.8**	−24.2	+102.6	−31.5	+209.8

Table 1 shows VSR-Place wins on every circuit and every metric against both CD legalizer configurations. Timing: VSR-Place 4.4 s avg, CD-standard 7.0 s, CD-scheduled 15.8 s.

4.3 Pareto characterization

Fig. 2 shows the tradeoff curve over $\lambda \in \{0, 0.25, 0.5, 1, 2, 4\}$. On 5/6 circuits the Pareto frontier includes an operating point that strictly dominates the unrepaired baseline.

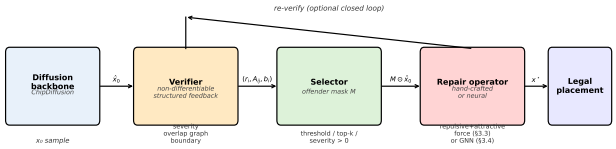

Figure 1: VSR-Place pipeline.

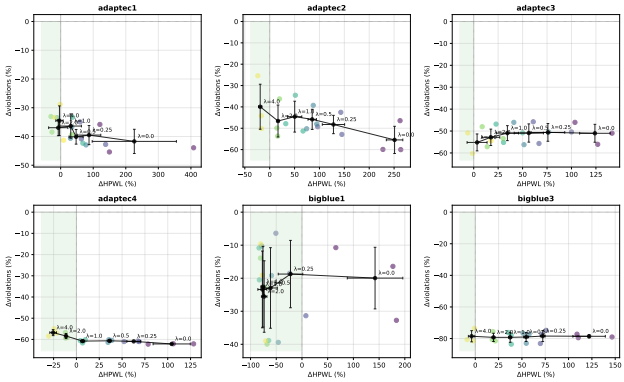

Figure 2: Pareto tradeoff between violation reduction and HPWL change over λ.

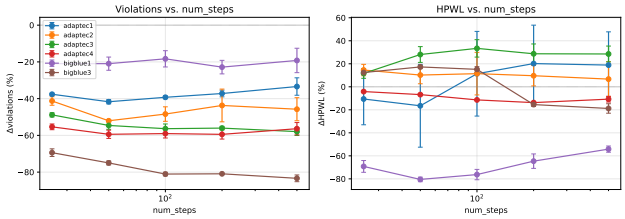

Figure 3: Ablation: violation and HPWL change vs. number of repair steps T.

4.4 Ablations

Fig. 3 shows that $T = 100$ is the elbow; diminishing returns beyond. Step size $\eta = 0.3$ is the robust middle ground. Selective vs. global repair is identical on ISPD2005 because raw diffusion drafts have near-universal violations.

5 Negative Result: Learned Operators

We also instantiated the repair operator as a 3-layer GATv2 (103K params) under six training regimes: synthetic teacher, real teacher, data augmentation, trajectory distillation, ground-truth target, and residual learning. Best val-loss reached 0.0037 (trajectory distillation) but no variant beat the hand-crafted baseline on ISPD2005. We attribute this to data scarcity: the placement benchmark yields 3–18 training samples after filter-

ing, orders of magnitude below typical GNN regimes.

6 Discussion

Why does CD-scheduled make HPWL worse? The scheduled config ramps up legality weight by 2× in the final 25% of steps and zeroes the HPWL term in the final 10%. The end phase aggressively relocates macros with no wirelength preservation. VSR-Place avoids this: the attractive term is never decoupled.

Limitations. bigblue2/bigblue4 OOM in the backbone (not VSR-Place). Standard-cell legalization is out of scope. We do not model routing congestion.

7 Conclusion

A simple non-differentiable verifier paired with a 50-line physics operator beats 5000-step constrained SGD on every ISPD2005 circuit we tested, on both legality and wirelength. Code and data released.